

What's New in Software Design Tools for Embedded Systems

Embedded design tools have become as important as microcontrollers for designers. These tools are getting more powerful, intuitive and flexible, allowing for new and better designs to be completed much faster

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Since hardware designs are now being churned out in large numbers, it has become challenging for designers to increase the adoption rate and at the same time keep pace with new developments. But thanks to the software tools which are maturing at par with the hardware developments and helping the design engineers to overcome these challenges.

If we see the latest developments, Silicon Valley is getting equipped with a new generation of design tools which are easy to adopt, scalable to design, economical to build and help reduce time-to-market the new innovations. Almost all silicon vendors have taken up more integrated approach in terms of coming up with their own development environment and support.

On software side, emphasis has been on performance analysis and better utilisation of CPU cores, while hardware developments are getting assisted better with the help of more and more simplified and integrated IDE approaches.

From simple PCB designing to complex FPGA and embedded board designs, we have a variety of software tools available with rich sets of features to help design easy, simplified and optimised designs. Let us see how these tools have matured in recent times.

Performance analysis support

On software front, if we look at the new features, focus has been particularly on performance analysis. There have been various enhancements and developments in software for better utilisation of CPU cores and system memory, improving overall efficiency of the embedded design tools.



For instance, TI's code composer studio has been enhanced immensely to help analyse performance of DSP as well as CPU cores. Mentor Graphics also announced next-generation Sourcery CodeBench and Sourcery Analyzer products to allow developers accelerate system debugging, including concurrent Linux applications, by quick and easy visualisation and analysis of complex software systems.

Also, the latest version of Intel System Studio includes an all-new analyser which analyses system and

SOC events. The new GUI displays the hardware events with call stacks, and results can be seen in source or assembly without any instrumentation.

Virtual JTAG probes

Software applications can be tested against IDEs and the hardware can be developed for dedicated logics simulated on the design tools, and you can easily get all the bugs out. But when the entire system is put together, interacting with the processor, there occur system-level problems which are very hard to capture.

Arun Mulpur, industry marketing manager, MathWorks, says, "The most basic question while talking about hardware/software co-design/debug is, where do I start?"

Whether to start with hardware and then write software, or first software then hardware? But actually doing both at the same time in the same environment is the solution. Simulink embraces this approach as a system-level simulation and implementation platform for both hardware and software."

The latest embedded design suites are addressing these system-level problems with new capabilities where software breakpoints can trigger hardware data capture and vice-versa. You can execute instruction traces or have your hardware breakpoints to halt application debugs to easily find out system-level problems, saving time and costs.

Mentor's Embedded Sourcery Codebench Virtual Environment for